DANDAN TAO

Google Scholar: https://scholar.google.com/citations?user=dWc_O5UAAAAJ
Email: ddantao@gmail.com

Education

2010 - 2015	Ph.D.	Department of Meteorology and Atmospheric Science, Pennsylvania State University (PSU), University Park, US
2006 - 2010	B.Sc.	Department of Atmospheric and Oceanic Sciences, Peking University, Beijing, China

Employment

2025 - present	Associate Professor, Nanjing University, China
2021 - 2024	Researcher, University of Bergen, Norway
2018 - 2020	Research scientist I, Colorado State University, US
2015 - 2018	Post-doctoral scholar, Pennsylvania State University, US

Publications

21. Tao, D., R. Nystrom, D. Chavas, and A. Avenas, 2026: A Fast Analytical Model for the Complete Radial Structure of Tropical Cyclone Low-Level Wind Field. *Geophysical Research Letters,* **53**, e2025GL118320.

20. Seland Graff, L., J. Tjiputra, A. Gjermundsen, A. Born, J. B. Debernard, H. Goelzer, Y.-C. He, P. M. Langebroek, A. Nummelin, D. Olivié, Ø. Seland, T. Storelvmo, M. Bentsen, C. Guo, A. Rosendahl, **D. Tao,** T. Toniazzo, C. Li, S. Outten, and M. Schulz, 2025: Sensitivity of winter Arctic amplification in NorESM2, *Earth Syst. Dynam*., **16**, 1671–1698.

19. Ito, K., Y. Miyamoto, C.-C. Wu, A. Didlake, J. Hlywiak, Y.-H. Huang, T. -K. Lai, L. Pattie, N. Qin, U. Shimada, **D. Tao**, Y. Yamada, J. A. Zhang, S. Kanada, and D. Herndon, 2025: Recent research and operational tools for improved understanding and diagnosis of tropical cyclone inner core structure. *Journal of the Meteorological Society of Japan*, **103**, 147-180.

18. Tao, D., C. Li, R. Davy, S. He, T. Spengler, C. Michel, and A. Rosendahl, 2025: Arctic-Atlantic cyclones: variability in thermodynamic characteristics, large-scale flow, and local impacts. *Geophysical Research Letters,* **52**, e2024GL111769

17. Boljka, L., I. Bethke, **D. Tao**, and C. Li, 2024: Upstream influence of midlatitude jet stream biases in boreal summer. *Atmospheric Science Letters*, **25**, e1272.

16. Rios-Berrios, R., P. Finocchio, J. Alland, X. Chen, M. Fisher, S. Stevenson, and **D. Tao**, 2023: A review of the Interactions between Tropical Cyclones and Environmental Vertical Wind Shear. *Journal of the Atmospheric Sciences*, **81**, 713–741**.**

15. Tao, D., R. Nystrom, and M. Bell, 2023: The quasi-linear absolute angular momentum slope of tropical cyclones under rapid intensification. *Geophysical Research Letters*, **50**, e2023GL104583.

14. Chen, X., C. M. Rozoff, R. F. Rogers, K. L. Corbosiero, **D. Tao**, J.-F. Gu, F. Judt, E. A. Hendricks, Y. Wang, M. M. Bell, D. P. Stern, K. D. Musgrave, J. A. Knaff, and J. Kaplan, 2023: Research advances on internal processes affecting tropical cyclone intensity change from 2018–2022. *Tropical Cyclone Research and Review*, **12**, 10-29.

13. Nam, C., M. Bell, and **D. Tao**, 2023: Bifurcation points for tropical cyclone genesis in sheared and dry environments. *Journal of the Atmospheric Sciences,* **80**, 2239–2259.

12. Casas, E., **D. Tao**, and M. Bell, 2023: A Simplified Intensity and Size Phase Space for Tropical Cyclone Evolution. *Journal of Geophysical Research - Atmospheres,* **128**, e2022JD037089.

11. Tao, D., P.J. van Leeuwen, M. Bell, and Y. Ying, 2022: Dynamics and predictability of tropical cyclone rapid intensification in ensemble simulations of Hurricane Patricia (2015). *Journal of Geophysical Research – Atmospheres,* **127**, e2021JD036079.

10. Tao, D., R. Rotunno, and M. Bell, 2020: Lilly's Model for Steady-State Tropical Cyclone Intensity and Structure. *Journal of the Atmospheric Sciences*, *77*, 3701-3720.

9. Tao, D., M. Bell, R. Rotunno, and P.J. Van Leeuwen, 2020: Why do the maximum intensities in modeled tropical cyclones vary under the same environmental conditions? *Geophysical Research Letters*, **47**, e2019GL085980.

8. Tao, D., K. Emanuel, F. Zhang, R. Rotunno, M. Bell, and R.G. Nystrom, 2019: Evaluation of the assumptions in the steady-state tropical cyclone self-stratified outflow using three-dimensional convection-allowing simulations. *Journal of the Atmospheric Sciences*, **76**, 2995-3009.

7. Tao, D., and F. Zhang, 2019: Evolution of dynamic and thermodynamic structures before and during rapid intensification of tropical cyclones: sensitivity to vertical wind shear. *Monthly Weather Review*, **147**, 1171-1191.

6. Liu, S., **D. Tao**, K. Zhao, M. Minamide, and F. Zhang, 2018: Dynamics and Predictability of Rapid Intensification of Super Typhoon Usagi (2013). *Journal of Geophysical Research - Atmospheres*, **123**, 7462-7481.

5. Cohen, Y., N. Harnik, E. Heifetz, D. S. Nolan, **D. Tao,** and F. Zhang, 2017: On the Violation of Gradient Wind Balance at the top of Tropical Cyclones. *Geophysical Research Letters*, **44**, 8017-8026.

4. Zhang, F., **D. Tao**, Y. Q. Sun, and J. D. Kepert, 2017: Dynamics and predictability of secondary eyewall formation in sheared tropical cyclones. *Journal of Advances in Modeling Earth Systems*, **9**, 89-112.

3. Tao, D., and F. Zhang, 2015: Effects of Vertical Wind Shear on the Predictability of Tropical Cyclones: Practical versus Intrinsic Limit. *Journal of Advances in Modeling Earth Systems,* **7**, 1534-1553.

2. Tao, D., and F. Zhang, 2014: Effect of environmental shear, sea-surface temperature and ambient moisture on the formation and predictability of tropical cyclones: an ensemble-mean perspective. *Journal of Advances in Modeling Earth Systems,* **6**, 384-404.

1. Zhang, F., and **D. Tao**, 2013: Effects of Vertical Wind Shear on the Predictability of Tropical Cyclones. *Journal of the Atmospheric Sciences*, **70**, 975-983.

Book chapters

1. Zhang, F., C. Melhauser, **D. Tao**, Y. Q. Sun, E. B. Munsell, Y. Weng and J. A. Sippel, 2016: Predictability of Severe Weather and Tropical Cyclones at the Mesoscales. Dynamics and Predictability of Large-scale, High-impact Weather and Climate Events (eds, J. Li, R. Swinbank, H. Volkert and R. Grotjahn). Cambridge University Press.

Teaching and Training Experience

2023 fall – 2024 spring	Pedagogical training	*MNPED660 Collegial Teaching and Learning in STEM Education, University of Bergen*
2021 fall	Co-lecturer	*GEOF213 Dynamics of the Atmosphere and Ocean* (*10 ECTS, equivalent to 3 credits*), Bachelor level, GFI, University of Bergen
2017 spring	Co-lecturer	*METEO521 Dynamic Meteorology (3 credits, equivalent to 10 ECTS)*, Ph.D. level, Department of Meteorology and Atmospheric Science, PSU
2011 fall	Teaching Assistant	*METEO526 Numerical Weather Prediction (3 credits, equivalent to 10 ECTS),* Ph.D. level, Department of Meteorology and Atmospheric Science, PSU

Service

2020 - present	Associate Editor, *Journal of the Atmospheric Sciences*
2021 - present	Reviewer of proposals for National Science Foundation (US)
2012 - present	Reviewer of research articles for *Journal of Atmospheric Science, Monthly Weather Review, Geophysical Research Letters, Journal of Geophysical Research – Atmospheres, npj Climate and Atmospheric Science, Journal of Applied Meteorology and Climatology, Journal of Climate, Journal of Meteorological Research, Atmospheric Research, Weather and Climate Dynamics, Atmospheric Chemistry & Physics, Advances in Atmospheric Sciences*

2026 Organizer and Co-chair of the sessions '*Machine Learning Applications to Tropical Cyclones*' and '*Tropical Cyclones: Forecasting and Predictability' in the 106th AMS annual meeting, US*

2021/2022/2024 Co-chair of *the session 'Tropical Cyclone Intensity Change in Moderate Vertical Wind Shear: Mechanisms, Observations, and Predictability' in the AMS's 34th/35th/36th Conference on Hurricanes and Tropical Meteorology*

2021 Committee member for *Max A. Eaton Student Prize in the 34th Conference on Hurricanes and Tropical Meteorology, American Meteorological Society, US*

2016 Coordinator of *the 7th EnKF Data Assimilation* workshop, State College, Pennsylvania, US

2016 Coordinator of *Symposium on Advanced Assimilation and Uncertainty Quantification in BigData Research for Weather, Climate and Earth System Monitoring and Prediction, State College, Pennsylvania, US*

2016 Coordinator of *2016 SPARC Gravity Wave Symposium, State College, Pennsylvania, US*

Honor and awards

2015 Group Achievement Award in leading the Penn State's participation of Hurricane and Severe Storm Sentinel, National Aeronautics and Space Administration (NASA) "*for outstanding achievements of the Hurricane and Severe Storm Sentinel (HS3) airborne mission to investigate the factors influencing hurricane intensity change.*"

2010 Pennsylvania State University Funds for Excellence in Graduate Recruitment

2009 Peking University Founder Scholarship

2009 Peking University Learning Excellent Prize

Research Highlights

2016 Earth & Space Science News: Wind Shear Measures Help Predict Tropical Cyclones (https://eos.org/research-spotlights/wind-shear-measures-help-predict-tropical-cyclones)